CE 310

Capstone Project Kickoff

Your Team, Your Dataset, Your Question

University of Arizona · Dr. Wooyoung Jung · Fall 2026

What the Capstone Is

For the first time this semester, your team picks its own dataset — any real-world civil or architectural engineering topic, from any public source.

Your team of 3 will choose one numeric predictor (X), one numeric outcome (Y), and one grouping variable that splits your data into two comparison groups, then apply the full CE 310 toolkit to it: regression, hypothesis testing, Monte Carlo simulation, and bootstrap confidence intervals.

Don't have a topic yet? See the dataset selection guide for what your data needs to support these methods, plus suggested public sources across transportation, water resources, structural, and energy topics. If you'd rather not search, 5 ready-made building-energy datasets (Hospital, PrimarySchool, LargeRetail, Warehouse, SmallHotel) are still available in the course files — a safe, proven option.

What You Will Build

Five sections in the capstone lab notebook, using generic X (predictor), Y (outcome), and GROUP variables mapped to your own dataset's columns:

A — EDA: histogram, scatter plot (colored by group), box plot by group · 3 answer codes

B — Regression: Model 1 (X only) and Model 2 (X + GROUP) · coefficients, R^2, bootstrap CI on the X slope, residual diagnostic plots · 5 answer codes

C — Hypothesis test: two-sample t-test comparing your two groups · 1 answer code + critical thinking

D — Uncertainty analysis: Monte Carlo simulation of your outcome under predictor uncertainty (same framework as Week 13) · 1 answer code + histogram

E — Executive summary memo: 3 paragraphs addressed to your dataset's client

Grading the Lab Notebook

This notebook is worth 40 points, graded individually and fully manually — there's no fixed "correct answer" key since every team's dataset is different; a preceptor reads your notebook and evaluates whether you applied each method correctly and interpreted your own results soundly.

Three Separate Deliverables

The capstone has three parts, graded independently — know all three deadlines now.

Deliverable	Points	Who submits	Due
Lab notebook (this document)	40	Individual — each teammate	Fri Dec 4, end of lab session
Capstone Report	40	Team — one submission	**Sat Dec 6, midnight**
Presentation	60	Team — presented live	Mon Dec 7 or Wed Dec 9

> ⚠️ The Capstone Report is a separate, team-written document — not a copy of your notebook's executive summary. See `the capstone report instructions` for what it requires.

Writing Your Client Brief

Before opening the notebook, your team writes its own client brief in your client brief — there's no instructor-provided brief this time. A good brief has:

- A named client/stakeholder who would realistically care about your dataset
- 2–3 specific questions the client wants answered
- The reason your dataset's two groups (GROUP_VALUE_A vs. GROUP_VALUE_B) matter to that client

The client brief includes a worked example (using one of the legacy building-energy datasets) showing the expected shape — use it as a template, not a source to copy from.

> ⚠️ Interpret every number in the context of YOUR dataset. A counterintuitive coefficient sign isn't necessarily wrong — it might be the most interesting finding in your analysis. Think through the mechanism before assuming it's an error.

Choosing Your Grouping Variable

Section C's hypothesis test and Section B's GROUP coefficient both depend on a well-chosen `GROUP_COL` — a categorical or binary column that splits your data into two meaningful comparison groups.

Good grouping variables: weekday vs. weekend, occupied vs. unoccupied, wet season vs. dry season, before vs. after a policy/design change, upstream vs. downstream — anything where you have a real hypothesis about why the two groups might differ.

Think ahead: does your grouping variable overlap with your predictor variable in any way (e.g., one season tends to have higher X values)? If so, that's a confounding relationship you'll need to discuss in Section C's critical-thinking prompt — not a problem to avoid, just something to explain.

Section D — Uncertainty Analysis

Section D uses the same Week 13 Monte Carlo framework, now applied generically:

```
X_UNCERTAINTY = 2.0   # <-- set this to fit YOUR predictor variable
delta_X = rng.normal(0, X_UNCERTAINTY, N)
Y_samples = (model2.params['Intercept']
             + (df['X'].mean() + delta_X) * model2.params['X']
             + df['GROUP'].mean() * model2.params['GROUP'])
```

The only thing that changes from team to team is `X_UNCERTAINTY` — how much your predictor might reasonably vary. Check `df['X'].std()` for a reference scale, then justify your choice in your Section E memo.

Writing the Executive Summary

Section E is worth 8 points and will be read as a professional deliverable. Your memo should:

- **Open with numbers:** ANSWER_A1 (row count), ANSWER_A2 (mean X), ANSWER_A3 (mean Y), ANSWER_B1 (X slope)
- **Explain the GROUP coefficient** in plain language for a non-statistician client. If it's counterintuitive, explain the practical reason.
- **Close with a recommendation:** one concrete action supported by the Monte Carlo P10–P90 range and your justified `X_UNCERTAINTY` choice

Avoid statistical jargon in the memo — write for your client, not your instructor.

Dec 7 and Dec 9: Presenting Your Findings

Each team presents a **4-minute** client briefing (+ 1 minute Q&A) on the last two class days (Mon Dec 7 and Wed Dec 9) — kept short because every team needs to fit into two standard 50-minute class sessions.

Slide structure (suggested):

1. Dataset & client — what's the data, who's the client, why does it matter
2. Regression results — Model 2 summary, what the GROUP coefficient means
3. Hypothesis test — is the group difference statistically significant? Cohen's d?
4. Uncertainty & recommendation — Monte Carlo histogram, P10/P90, actionable recommendation

Looking Ahead to the Week 16 Debrief

Since every team's topic is different this time, the Week 16 debrief focuses on **cross-topic methodology**, not shared numbers: who found a counterintuitive coefficient sign and how did they explain it? Whose R^2 was strongest, and why might that be (data quality vs. inherent variability in that domain)?

Checklist Before Starting

1. **Pick your dataset** as a team — see the dataset selection guide if you need help, or use one of the 5 legacy building-energy CSVs if you'd rather not search
2. **Upload your CSV to Colab**: Files icon → Upload to session storage
3. **Write your client brief** in your client brief
4. **Fill in Cell 3** of the capstone lab notebook: dataset name/source and the four column-role variables
5. Run cells top-to-bottom. All `ANSWER_X` variables are computed in order.

> Since every team's dataset is different, there's no "expected value" to check your numbers against. Instead, sanity-check them yourself: does your R^2 make sense given how noisy you'd expect your outcome to be? Does your GROUP coefficient's sign match your own physical reasoning? If not, that's worth digging into before you submit.